Prob. 9. (2 & T. Ch. 8. 2.1)

Consider the initial value Prob-m (IVP) for the one-dim-l wave eq-n:

(1) $$u_{xx} - u_{tt} = 0, \quad x \in \mathbb{R}, \quad t > 0$$

(2) $$u(x,0) = \varphi(x), \quad u_t(x,0) = \psi(x), \quad x \in \mathbb{R}$$

The sol-n is given by the d'Alambert's sol-n:

$$u(x,t) = \frac{1}{2}\left[\varphi(x+t) + \varphi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(\tau)\,d\tau.$$

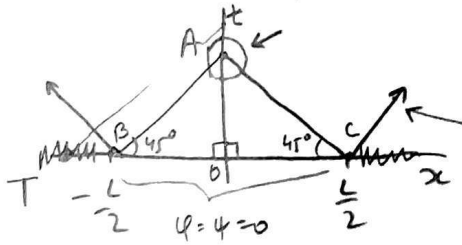
(a) Suppose that $\varphi = \psi = 0$ in some open interval of length L. Up to what time t can you be sure that $u = 0$ at the center of the interval (i.e $u(L/2, t_0) = 0$ $t_0 = ?$)?

Hint: Since you can determine where the coordin. syst is, make your life easier & put $x = 0$ at the center of the interval so that the interval where $\varphi = \psi = 0$ is $-L/2 < x < L/2$.

Sol-n: [Preamble].

Let $\varphi = \psi = 0$ on $\left(-\frac{L}{2}, \frac{L}{2}\right)$ as follows pictorially:

(3)



The direction that information propogates is $\xi = \langle 1, 1\rangle$ & $\langle -1, 1\rangle$.

(*) Since the direction of information makes 45° with x axes, $\varphi = \psi = 0$ makes an influence on $u(x,t)$ until $t = A$, after that other pnts outside of $(-L/2, L/2)$ (like T in the pic.) can influence $u(x,t)$ making it possibly nonzero. So

(4) $$u(x,t) = 0 \quad \text{until} \quad t = A,$$

and by geometry $AO = OB = OC = \frac{L}{2}$ (right Δ), so until

(5) $$\boxed{t = \frac{L}{2}}, \quad u(x,t) = 0.$$

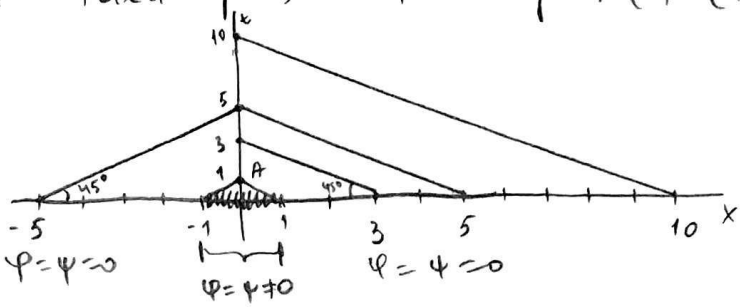
(b) Suppose that $\varphi = \psi = 0$ outside of the interval $|x| \le 1$. Up to what time can you be sure that $u = 0$ at (i) $x = 3$, (ii) $x = 10$, & (iii) $x = -5$.

Sol-n

[Preamble]

Using the similar idea presented in part (a) (*),

(b)



(**) Since $\varphi \neq 0$ & $\psi \neq 0$ on $|x| \le 1$, these parameters $\varphi, \psi \neq 0$ can influence $u(x,t)$ until $t = 1$, meaning after that they do NOT have any influence on $u(x,t)$ to be nonzero. Hence if $t > 1$, then $u(x,t) = 0$. Then

(i) $x = 3$, $u(x,t) = 0$ for $t > 1$ &

(ii) $x = 10$, $u(x,t) = 0$ for all $t > 1$.

(iii) $x = -5$, $u(x,t) = 0$ for all $t > 1$.